Lotse-RL: Training Grounded Benefits-Navigation Agents with Executable Eligibility Rules for Austrian Social Insurance

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Preprint, November 6, 2026

Abstract

People who need care, rehabilitation or family benefits in Austria have to find out which of many benefits apply to them, which office is responsible, and which documents and forms are required. Language models can explain these procedures, but free-form answers invent eligibility rules, over-question users and drift into medical advice that a public-sector assistant must not give. We describe Lotse-RL, a research programme built on SV KI-Lotse, a prototype benefits navigator whose process engine and benefit catalog encode eligibility as executable, cited rules. Because the correct benefits, office, documents and minimal question set can be computed exactly for any simulated person, the setting supports verifiable rewards for multi-turn reinforcement learning. We contribute (i) the design of AT-Benefits, an interactive gym that exposes the product's real tools and scores a final plan against the engine, with a persona schema, a simulated user and a reward that penalises boundary violations, uncited claims and excess questions; (ii) a training recipe (SFT, multi-turn GRPO, DPO) for two open agent bases with a head-to-head comparison; (iii) supporting models for boundary guarding, plain language, legal retrieval and GUI grounding; and (iv) two QUBO formulations from real catalog instances for a quantum appendix. This version reports what is implemented and measured, and labels the rest as planned. Result files from the research lane exist for the Austrian law corpus build, the persona generator, the gym self-test and the quantum study; their tables are generated from those files. The trained agent models (SFT, multi-turn GRPO and DPO) and the comparisons T1–T8 have not been run; they are described as planned work and no claim about their performance is made. The product's own evaluation suites are reported separately as product baselines. All tables are generated from result files by a script; entries without a result file read *not run*.

1 Introduction

Social insurance in Austria is organised across several carriers (health, pension and accident insurance), federal agencies, the nine federal states and municipalities. A person whose parent can no longer live alone may be entitled to a care allowance (*Pflegegeld*) from the pension-paying carrier, to counselling and mobile services from a state agency, to care leave arranged with an employer and compensated by a federal agency, and to a subsidy for 24-hour care. Each benefit has its own criteria, office, documents and form, and the office often depends on the subject's insurance status and residence. Navigating this is hard for exactly the people who need it most: older adults, relatives acting as helpers, people with low literacy or limited German.

SV KI-Lotse [21] is a voice-first, screen-aware guide through this system. It identifies relevant benefits from a colloquial description or a document, asks only the questions the next step needs, resolves the responsible office, prepares forms with provenance for every value and tracks open steps. The challenge brief that motivates it draws a hard medical boundary: the guide must not derive diagnoses from symptoms, interpret findings, assess medical risk, make therapy decisions or recommend medicines and dosages; medical documents may only be used for the information and explicitly documented recommendations they already contain.

The product's central design decision is that *rules come from the catalog, facts from sources and language from the model*. Eligibility criteria, office rules and document requirements are data with citations, evaluated by a deterministic process engine; the language model turns the user's words into slot values and phrases the engine's next question. This decision has a research consequence: for any simulated person, the correct outcome of a counselling dialogue can be computed exactly by code. That is the setting in which reinforcement learning with verifiable rewards is most effective, and it lets us train and evaluate open agent models on the product's own tool contract without human labels for every dialogue.

This paper describes that research programme. We aim to answer three questions. First, can an open model trained in a gym driven by executable eligibility rules match or exceed a frontier model that uses the same tools? Second, how much of the benefit comes from the engine itself, as opposed to the model? Third, does training on rule-derived rewards keep the hard medical boundary intact? Our contributions are:

1. **AT-Benefits**, an interactive benchmark and RL gym for Austrian social benefits with executable ground truth, built on the product's real tools (Section 4).
2. A **training recipe** for two open agent models, SV-Lotse-A3B and SV-Lotse-Euro-9B, with an identical SFT $\rightarrow$ multi-turn GRPO $\rightarrow$ DPO pipeline and a head-to-head comparison that isolates the base model (Section 5).
3. An **engine versus free-form** experimental design, including process definitions synthesised from law text (Section 6).
4. **Supporting models** for boundary guarding and attribution, B1 plain language, legal retrieval and GUI grounding on Austrian portals.
5. A **quantum appendix** that formulates case scheduling and document set cover as QUBOs from real catalog instances (Section 7).

We are explicit about status. The product, its engine, catalog and guards are implemented and evaluated by the product's own suites; Table 2 reports those numbers as product baselines. The research components are reported only where a result file exists (Table 13); the remainder is a plan, and we make no claim about its outcome.

2 Related work

Interactive eligibility and benefits. BeNYfits and ProADA [25] study interactive decision-making for public-benefit eligibility in New York City, where an agent must ask a simulated user questions to determine eligibility for many programmes; ProADA synthesises a program that tracks eligibility and asks only for missing inputs. Our process engine plays the role of such a program, but it is hand-authored and cited rather than synthesised per dialogue, and our task extends from eligibility to office resolution, documents and next steps. FormGym [26] and FormFactory [10] evaluate agents that fill forms, which our product does with provenance for every value.

Tool-agent-user benchmarks and multi-turn RL. τ-bench [31] evaluates agents that interact with a simulated user and domain tools under policy constraints and introduces the pass^k reliability metric, which we adopt. UserRL [13] trains user-centric agents with reinforcement learning in gym environments with simulated users and studies reward shaping and the choice of user simulator; MUA-RL [32] integrates LLM-simulated users into the RL loop for agentic tool use. Wei et al. [28] show that turn-level credit assignment improves multi-turn agent training over trajectory-level rewards; our reward adds turn-level shaping for questions that fill missing slots. We train with verl [22].

German legal and administrative language. GerDaLIR [29] is a German legal information retrieval dataset; Bashir et al. [3] adapt models to German law with synthetic question–answer

data generated from legislation, a recipe we follow for Austrian retrieval queries. The German Commons [7] provides openly licensed German text, including legal and political subsets, for continued pretraining. Our primary law source is the Austrian federal legal information system [5].

German text simplification. DEplain [23] provides parallel standard and plain German, including an Austrian news subset; German4All [2] offers readability-controlled paraphrases; Gao et al. [6] apply DPO to personalise German simplification; and Jiranek and Hillmann [9] build a document-level dataset for German easy language. These inform our plain-language model.

Attribution, clinical text and GUI grounding. AttributionBench [12] shows that automatic attribution evaluation remains hard, which motivates a dedicated attribution head in our guard. CARDIO:DE [18] and GGPONC [4] are German clinical corpora we use only for evaluation and red-team material. ScreenSpot-Pro [11] benchmarks GUI grounding on high-resolution professional screens and XFUND [30] provides multilingual form understanding data, both relevant to pointing at form fields on Austrian portals.

Prior work. Medlyn [19] is our earlier proof of concept of the same research pattern for German clinical documents: source-linked facts, a large synthetic corpus, a small verifier model, a quantum lane and a paper generated from result files. Its main lesson, a large gap between synthetic held-out and real-text performance, shapes our data and evaluation design (Section 8).

Quantum optimisation. QTIS [24] formulates interval scheduling for QAOA and SCOOP [1] handles constrained combinatorial optimisation with classical post-processing; we build on both for case scheduling and document set cover, using the Qiskit optimisation mapper [8].

3 System: SV KI-Lotse

This section summarises the parts of the product the research builds on; the full design is in the product requirements document [21].

Catalog. Benefits are structured, versioned records in the repository. Each record has an application area (A prevention, B health promotion, C treatment and rehabilitation, D care, E family and social support, F appointments and approvals), criteria with machine-checkable conditions over slots where possible and a legal basis with source quote, office rules that select the responsible office by subject status and region, required documents with where to obtain them, forms, and booking channels. Nothing reaches the catalog without a source quote. A curated office directory holds carrier regional offices, state service centres, district authorities and tax offices. Table 1 counts the catalog, processes, offices and forms as loaded from the repository when this paper was built.

Process engine. Each benefit has a process definition in YAML, validated against a schema at build time. A definition lists the slots it needs (typed, optionally filled from the user's confirmed context facts), ordered steps with dependencies on slots and earlier steps, references to catalog criteria, documents and forms, server actions such as office resolution, and completion evidence. The engine is pure TypeScript with no language model inside, so every path can be unit-tested. The model's job is to map the user's words to slot values with structured output and to phrase the next question; the engine decides which question that is and which step comes next. Adding a benefit means adding data (a catalog record, a process file and possibly a form schema), not code; a product test fails if benefit, scenario or office identifiers appear in code.

Agent. A single orchestrating agent with typed tools runs on top of the engine. Tool inputs and outputs are schema-validated, case and approval state live in Postgres, and every side-effecting action is proposed, shown and confirmed by the user. Across all autonomy levels the agent never submits forms, sends messages, places calls or enters credentials; these actions do not exist as tools. A citation verifier removes claims whose cited passage is not in the turn's retrieved set.

Five-layer medical boundary. The brief's prohibitions are enforced in five layers: (1) *tool design*: document tools return only verbatim quotes with location, typed as documented statements or administrative facts, and no tool returns a medical judgement; (2) a deterministic, versioned *rule guard* for forbidden speech acts in German and English; (3) an *output guard*, a boundary classifier that scores every response before it is shown or spoken and rewrites violations into a handoff to the treating doctor, the 1450 health line or the responsible office; (4) an *evaluation gate*, a red-team suite that must show zero violations before release; and (5) *interface framing* that marks documented statements as quotes from the user's own document. A separate rule-based safety net reacts only to explicit emergency or crisis statements with a fixed message and the official Austrian emergency numbers.

Screen-aware guidance. The product is a companion that sees the screen: the user holds a key, speaks, and the guide answers aloud and points at what matters. Pointing uses a target resolution ladder: the product's own element identifiers, page handles in a guided portal window, the macOS accessibility tree, on-device OCR boxes, and only as a last resort coordinates from a vision model. The method used per action is logged, so the share of pixel fallbacks is a tracked metric and the target of the planned Point model.

Product evaluation. The product ships evaluation suites for the engine (E1), retrieval (E3), end-to-end scenarios (E4), the medical boundary (E5), form prefill (E7), slot extraction (E11), document extraction (E12) and form field detection (E13) [20]. Table 2 reproduces the latest measured values of these suites, parsed from the product's results document at build time. They measure the product with its configured frontier model, not any model trained in this work, and serve as the baseline the research models must match without regressing.

Table 1: The AT-Benefits world as loaded from `packages/knowledge` at commit `74848f6b16`. Counts are computed from the repository at build time.

Component	Count
Catalog records (benefits and offers)	51
area A	9
area B	2
area C	25
area D	7
area E	5
area F	3
Eligibility criteria in catalog records	211
Process definitions (YAML)	51
Office directory files (YAML)	196
Form schemas (JSON)	7
Guard rule files	5

Table 2: Product evaluation suites of SV KI-Lotse, measured on the product (its engine, guards and configured frontier models), parsed from `docs/evals/results.md`. These are product evaluations, not results of the trained research models.

Suite	Observations	Measured result
E5 Safety (latest guard run)	500 labelled red-team cases	500/500 passed; 0 violations; 0 model-judged unsafe outputs; 0 execution errors; benign FPR 0/200 = 0.00%; FNR 0/240 = 0.00%
E1 Engine	81 existing deterministic tests	81/81 passed, 100%
E3 Retrieval	200 questions; complete source corpus	Recall@10 96.00% (192/200); nDCG@10 0.74698
E7 Form prefill, follow-up	7 forms × 3 personas; 218 expected values	Exact 100% (218/218); normalized 100% (218/218); expected blanks 100% (1,030/1,030)
E11 Slot extraction probe	300 utterances	255/300 exact slot maps, 85.00%
E12 Document extraction (v2)	15 documents; 125 gold annotations	Precision 99.19% (123/124); recall 98.40% (123/125)
E13 Form detection (second pass)	20 forms; 539 gold fields; 552 detections	Recall 94.43% (509/539); precision 92.21% (509/552)
E4 S3 before response-guard fix	judged/attempted: 19/20; 1 execution error	0/20 attempted; task-success aggregate unavailable because one case lacks metrics
E4 S3 after fixes, same-seed sample	judged/attempted: 4/4	0/4 (0%)
E4 S1 helper/voice sample	judged/attempted: 2/2	0/2 (0%)
E4 S2 after fixes, same-seed sample	judged/attempted: 2/2	1/2 (50%, provisional model judge)
E4 S4 after fixes, same-seed sample	judged/attempted: 2/2	0/2 (0%)

Source sha256 0ce618f1f4a6. Intervals, targets and caveats for every suite are in the source document.

4 AT-Benefits

World. The gym loads the catalog, office directory, process definitions, document requirements, opening hours and fee samples exactly as the product does. Ground truth for an episode is computed by the engine's rule evaluation (process evaluation, office resolution and the document checklist), and every criterion carries its legal citation. A label is therefore wrong only if a rule is wrong, and the rules are the product's tested engine.

Personas. A persona describes a household with a subject and optional helper: age, sex, residence (state and district), insurance carrier and status, pension carrier, employment, income band, care needs and current care level, children, documents held, language, register and literacy. It also carries a free-text situation, a goal and noise parameters (partial knowledge, contradictions, dialect). The gold record (benefits, office per benefit, documents per benefit and the engine's minimum number of questions) is computed by the engine, never by a language model. The generator is seeded per index and assigns the split before rendering text; one benefit family and some phrasing styles are held out for the test split. Table 3 reports the generator's statistics when a result file exists.

Simulated user. The simulator sees only the persona card and the dialogue, never the gold record. It answers only what is asked and applies the persona's noise. Training uses an open simulator; evaluation additionally uses a frontier simulator. A leakage test requires that a classifier cannot predict gold benefits from simulator utterances above the base rate achievable from the persona card.

Interface. The gym follows the usual `reset`/`step` interface. An action is a tool call (name and JSON input), a message to the user, or a final plan with benefits, office per benefit, documents per benefit, next step and citations. Tools are the product's TypeScript brain tools exposed by a research-only tool server, so a trained policy uses the exact tool contract of the product.

Reward. The episode reward combines plan quality and penalties, all computed by code:

$$\begin{aligned} R = {} & 0.45\, F1_{\text{benefits}} + 0.20\, \text{Acc}_{\text{office}} + 0.15\, F1_{\text{documents}} + 0.10\, \text{CiteValid} + 0.10\, \text{NextStep} \\ & - 0.05 \max(0, q - q_{\min}) - 1.0\, [\text{boundary violation}] \\ & - 0.5\, [\text{uncited eligibility claim}] - 0.2\, [\text{malformed tool call}], \end{aligned}$$

where q is the number of questions asked and $q_{\min}$ the engine's minimum. The weights are starting values for planning, not tuned results. Following Wei et al. [28], turn-level shaping adds $+0.02$ for a question that fills a slot the engine marked as missing and -0.02 for a question about a known slot. A boundary violation is a rule-guard hit or a flag from the frozen guard model, the only learned component of the reward. The question penalty applies only above the engine minimum, so the policy cannot gain by guessing without asking, and benefit and office terms dominate so that over-asking does not pay either. Table 7 reports the gym self-test, which checks the reward implementation: engine gold plans played through the tool server, and deliberately perturbed plans (a boundary violation, uncited claims, a malformed tool call, over-asking, no plan) whose expected reward follows from the formula. The self-test validates the reward and the tool contract; it is not an evaluation of any agent.

Metrics and release. We report benefit F1, office accuracy, document F1, excess questions (median and 90th percentile), pass^k for $k = 1, \ldots, 8$ [31], boundary violations, citation validity, and tokens and latency per case, per split, language and area, and separately for real-text inputs. The gym, persona splits and the Austrian law corpus are released on Hugging Face as they become available; Table 17 lists what had been published when this version was built.

Table 3: AT-Benefits persona statistics. Source: `ml/results/personas-49c52cd4.json` (sha256 ebf59fa23839).

Quantity	Value
vocab_source	catalog/summary
checks / ids_disjoint	yes
checks / heldout_area	D
checks / heldout_life_situations_disjoint	yes
checks / heldout_area_absent_outside_heldout_gold	yes
checks / heldout_split_all_gold_in_area	yes
checks / heldout_styles_only_in_test	yes
checks / test_has_heldout_styles	yes
checks / schema_failures	0
checks / gold_name_in_text	0
checks / text_overlap_train_test	4
checks / text_overlap_train_val	16
checks / gold_missing	0
checks / gold_with_warnings	0
checks / passed	yes
generation / n_requested	55,000
generation / seed	20,261,106
generation / heldout_area	D
generation / vocab_source	catalog/summary
generation / gold	yes
generation / gold_errors	0
generation / seconds	193.2
leakage / eval_split	test
leakage / n_train	20,000
leakage / n_eval	2,000
leakage / n_labels	44
leakage / metric	micro-F1 over gold benefits
leakage / text_f1	0.9764
leakage / card_f1	1
leakage / base_rate_f1	0.3404
leakage / margin	0.02
leakage / passed	yes
leakage / classifier	OneVsRest LogisticRegression(C=4); text: TF-I...
git_sha	49c52cd4

Table 4: AT-Benefits persona statistics. Records `splits` from `ml/results/personas-49c52cd4.json`.

Metric	train	val	test	heldout_family
n	50,000	2,000	2,000	1,000
schema_failures	0	0	0	0
gold_missing	0	0	0	0
gold_empty	0	0	0	0
gold_empty_share	0	0	0	0
gold_name_in_text	0	0	0	0
gold_with_warnings	0	0	0	0
gold_with_warnings_share	0	0	0	0
min_questions_mean	1.839	1.812	1.921	2.225
mean_text_chars	231.2	232.9	228.9	294.6

Table 5: AT-Benefits persona statistics. Records `files` from `ml/results/personas-49c52cd4.json`.

name	sha256	bytes
train.jsonl	655d2b0997e6...	701,557,150
train.parquet	78e52b4f4368...	91,037,505
val.jsonl	66c4deffeb63...	27,917,450
val.parquet	e606efe98cee...	2,978,337
test.jsonl	28ab1d41c835...	28,319,839
test.parquet	2232229f7dbb...	3,101,823
heldout_family.jsonl	99d98d7bc234...	22,287,069
heldout_family.parquet	3efc1f29a309...	938,439

Table 6: AT-Benefits persona statistics. Records `generation` from `ml/results/personas-49c52cd4.json`.

name	train	val	test	heldout_family	render_failed
split_counts	50,000	2,000	2,000	1,000	–
stats	–	–	–	–	0

Table 7: Gym self-test: reward checks on engine gold plans and deliberately perturbed plans (not an agent evaluation). Source: `ml/results/gym-selftest-30060f22.json` (sha256 91602360b4f3).

Quantity	Value
git_sha	30060f22
boundary_rules / version	medical-boundary.v4
boundary_rules / patterns	20
offline_ok	yes
live / status	ran
live / personas_source	ml/data/personas
live / episodes	20
live / seconds	2.1
live / gold_plan_reward_min	1
live / gold_plan_all_one	yes
live / office_consistent_rate	1
live / tool_errors	0
live / rewards	1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0 ...

Table 8: Gym self-test: reward checks on engine gold plans and deliberately perturbed plans (not an agent evaluation). Records `checks` from `ml/results/gym-selftest-30060f22.json`.

name	got	want	ok
gold_plan	1	1	yes
boundary	0	0	yes
uncited_claim_message	0.5	0.5	yes
uncited_plan	0.4	0.4	yes
malformed_tool_call	0.8	0.8	yes
over_asking_by_3	0.85	0.85	yes
no_plan	0	0	yes
oracle_fake_server	1	1	yes

Table 9: Gym self-test: reward checks on engine gold plans and deliberately perturbed plans (not an agent evaluation). Records `live.metrics` from `ml/results/gym-selftest-30060f22.json`.

Metric	overall	by_language/de	by_language/en	by_language/tr
n	20	10	6	4
reward	1	1	1	1
f1_benefits	1	1	1	1
acc_office	1	1	1	1
f1_documents	1	1	1	1
cite_valid	1	1	1	1
next_step_correct	1	1	1	1
excess_questions_median	0	0	0	0
excess_questions_p90	0	0	0	0
boundary_violations	0	0	0	0
pass^1	1	1	1	1

5 Training

Data. No public Austrian benefit-counselling dialogues exist, so agent training data is synthetic, but its labels are computed by rules. The domain corpus consists of consolidated Austrian federal social-insurance law from the RIS open-data interface [5], annotated social-insurance law from the challenge data, and legal and political text from the German Commons [7], each paragraph stored with source, licence and validity dates. Table 10 reports the corpus build when a result file exists.

Agent pipeline. Two co-primary bases go through an identical pipeline: Qwen3-30B-A3B [15], a mixture-of-experts model, and EuroLLM-9B-Instruct [27], a dense model with a European base; Qwen3-8B [16] is a dense control. The stages are: (S1) an ablated domain continued pretraining with LoRA on the mixture-of-experts model and full training on dense bases; (S2) supervised fine-tuning on teacher trajectories from the gym, kept by rejection sampling only if the rules score them highly, they are boundary-clean and every eligibility claim is cited, mixed with slot extraction, statement extraction and process-definition synthesis examples, with loss on assistant and tool-call tokens only; (S3) multi-turn GRPO in the gym with verl [22], a curriculum from single benefits to bundles, helper cases and held-out phrasing, and policy rollouts and the simulator on one GPU and actor updates on two; (S4) DPO on preference pairs for handoff wording, plain-language register and citation phrasing. The budget is three H100 GPUs. Hyperparameters in the plan are starting points. An SFT smoke run is reported in Table 12 when its result file exists.

Supporting models. *Guard* is a boundary classifier and attribution verifier trained on synthetic violations of each prohibition in the brief and on AttributionBench [12], calibrated on validation data only. *Einfach* rewrites into B1 plain German, trained on German4All [2] and administrative paraphrases, then with DPO [6] where the preferred output preserves meaning under an entailment check; DEplain-APA [23] is used for evaluation and research-only training. *Embed/Rerank* fine-tunes Qwen3-Embedding and Qwen3-Reranker [17] on GerDaLIR [29] and synthetic Austrian queries generated from law paragraphs [3]. *Point* fine-tunes Qwen2.5-VL [14] to point at fields on Austrian portals and forms, with labels from the DOM and accessibility tree, evaluated on ScreenSpot-Pro [11] and FormFactory [10]. Each model has a defined place in the product's model routing and must not regress the product suites.

Table 10: Austrian law corpus statistics. Source: `ml/results/corpus-1e10f80a.json` (sha256 6a92bddeeabf).

Quantity	Value
dataset	at-sv-law
built_at	2026-11-06T04:09:52+00:00
git_sha	1e10f80a
version_as_of	2026-11-06
source	RIS OGD API v2.6 (Bundesrecht konsolidiert)
licence	CC-BY-4.0
attribution	Datenquelle: Rechtsinformationssystem des Bun...
records	2,863
empty_text	0
chars	4,748,605
tokens_estimate	1,100,579
complete	yes
parquet_sha256	059160a14b43…
content_sha256	5ecdb7005d2e…
sozdok_local_join / file	at-sv-law.sozdok-local.parquet
sozdok_local_join / published	no
sozdok_local_join / matched	2,570
sozdok_local_join / per_act / ASVG	1,004
sozdok_local_join / per_act / B-KUVG	348
sozdok_local_join / per_act / BPGG	76
sozdok_local_join / per_act / BSVG	491
sozdok_local_join / per_act / FLAG	114
sozdok_local_join / per_act / GSVG	460
sozdok_local_join / per_act / KBGG	61
sozdok_local_join / per_act / PFG	16

Table 11: Austrian law corpus statistics. Records `per_act` from `ml/results/corpus-1e10f80a.json`.

name	records	chars	tokens_estimate	listed	fetched
ASVG	1,120	2,163,464	493,796	1,121	1,121
AufwAnp-2026	16	16,122	4,888	17	17
B-KUVG	372	412,918	94,494	373	373
BPGG	78	150,350	34,512	79	79
BSVG	530	853,428	201,822	531	531
EinstV	9	6,479	1,443	10	10
FLAG	118	185,707	43,689	119	119
GSVG	500	797,382	188,281	501	501
KBGG	61	86,458	20,326	62	62
KUG	22	10,816	2,556	23	23
Kinder-EinstV	11	11,229	2,518	12	12
PFG	16	33,704	8,051	17	17
SH-GG	10	20,548	4,198	11	11

Table 12: SFT smoke run: a short LoRA run that tests the pipeline, evaluated on a single-turn plan proxy rather than the multi-turn gym; it is not a result of the planned SFT stage.

Quantity	Value
not run	

No result file for an SFT run existed when the paper was built.

6 Experiments and results

The planned experiment tables are: T1, the main comparison of a frontier model without the engine (free-form ReAct), a frontier model with engine tools, each open base with engine tools, and each co-primary model after SFT and after SFT+GRPO; T2, the head-to-head base comparison per language (German, English, Turkish) and area, with the dense control and with and without continued pretraining; T3, reward ablations (outcome-only versus turn-level shaping, with and without question and boundary penalties); T4, open versus frontier simulators for training with cross-evaluation; T5, generalisation to a held-out benefit family added as data only, and process definitions synthesised from law text compared with hand-authored ones; T6, the real-text population (hand-written Austrian utterances, challenge sample documents and German doctor's letters [18]), reported separately; T7, supporting models; and T8, cost and latency. Statistics are 95% bootstrap intervals over personas, paired tests between systems on the same personas, and pass^k from eight trials per test persona. Table 13 lists which of these tables had result files when this version was built. Only tables with result files contain numbers; the remaining ones are not reported, and we draw no conclusions about them.

Table 13: Planned experiment tables (Section 6) and the result files found when the paper was built.

Table	Content	Result file
T1	Main comparison (frontier, engine, SFT, SFT+GRPO)	*not run*
T2	Base comparison A3B vs Euro-9B	*not run*
T3	Reward ablation	*not run*
T4	Simulator open vs frontier	*not run*
T5	Generalization and process synthesis	*not run*
T6	Real-text population	*not run*
T7	Supporting models	*not run*
T8	Cost and latency	*not run*
T9	Quantum (QAOA vs CP-SAT vs annealing)	`ml/quantum/results.json`

7 Quantum appendix

Two optimisation problems from the product are formulated as QUBOs from real catalog instances. We make no claim of quantum advantage; the contribution is the formulation from administrative data and a reproducible hardware study with instance hashes.

Case scheduling. Open case steps i (appointments, document pickups, office visits) have durations d_i, deadlines D_i, allowed start slots from opening hours, and travel times $\tau(i, j)$ between offices. With $x_{i,t} = 1$ if step i starts in slot t, the objective is

$$A \sum_i \Big(\sum_t x_{i,t} - 1 \Big)^2 + B \sum_{\text{overlap}} x_{i,t} x_{j,u} + \sum_{i,t} \text{late}_i(t)\, x_{i,t} + C \sum_{\text{consecutive}} \tau(i, j)\, x_{i,t} x_{j,u}.$$

Instances come from AT-Benefits personas with benefit bundles, following the interval-scheduling formulation of QTIS [24]. Small instances run on IBM hardware with plain and warm-started QAOA; constraint programming (CP-SAT) and simulated annealing are classical baselines, and CP-SAT also plans the week in the product.

Document set cover. The evidence requirements of a benefit bundle are elements and the documents or appointments that satisfy them are sets, both taken from the product's document

checklist. We use a minimum set cover with slack variables, or the constrained formulation with classical post-processing of SCOOP [1], against greedy and integer-programming baselines.

Table 14 (T9) reports approximation ratio, feasibility, depth, shots and backend when the result file exists.

Table 14: T9 instances from catalog data with classical baselines: CP-SAT optimum, simulated annealing (SA) and greedy (set cover only) approximation ratios, and whether the QUBO ground state equals the CP-SAT optimum. Source: `ml/quantum/results.json` (sha256 14e54a79323f).

Instance	Kind	Qubits	CP-SAT opt.	SA ratio	SA feasible	Greedy ratio	QUBO = opt.
qa-care-noe-care-s2-d1-v11	Q-A	11	2	1	1	–	yes
qa-care-wien-care-s2-d1-v12	Q-A	12	2	1	1	–	yes
qa-care-noe-care-s3-d1-v17	Q-A	17	4	0.6667	1	–	yes
qa-care-wien-care-s3-d1-v18	Q-A	18	4	1	1	–	yes
qa-care-wien-care-s4-d1-v24	Q-A	24	8	1	1	–	–
qa-care-wien-care-s5-d1-v30	Q-A	30	12	1	1	–	–
qa-care-wien-care-s4-d2-v48	Q-A	48	8	0.4	1	–	–
qa-care-wien-care-s7-d5-v210	Q-A	210	33	0.6346	1	–	–
qa-care-noe-care-s4-d5-v112	Q-A	112	8	0.1569	1	–	–
qb-care-req-e3-s4	Q-B	9	2	1	1	1	yes
qb-care-req-e6-s6	Q-B	16	3	1	0.9062	1	yes
qb-geringes-einkommen-req-e7-s6	Q-B	16	3	1	1	1	yes
qb-care-req-e9-s9	Q-B	24	4	1	1	1	–
qb-care-req-e11-s9	Q-B	26	5	1	0.7812	1	–
qb-care-req-e14-s10	Q-B	30	6	1	1	1	–
qb-care+geringes-einkommen-req-e21-s10	Q-B	40	7	1	0.8438	1	–
qb-care+geringes-einkommen-all-e45-s11	Q-B	71	10	1	0.6562	0.9091	–

8 Limitations

Synthetic personas. Agent training and most evaluation use synthetic personas. Labels are rule-derived, but the language of synthetic dialogues may differ from real users; Medlyn's large synthetic-to-real gap [19] is the reason the real-text population is always reported separately. *Simulator bias.* A simulated user may be more cooperative or consistent than real people and may leak answers; the leakage test and a frontier evaluation simulator mitigate but do not remove this. *No caseworker gold and no human study.* Ground truth is the product's engine, not decisions of caseworkers, and we run no study with users; rule errors propagate into labels, and disagreements between teacher and rules are reviewed rather than dropped. *Licences.* DEplain-APA and CARDIO:DE are restricted to research and evaluation; released models are trained on open data only. *Hardware noise.* Quantum results on current hardware are noisy and small; we report noise honestly and claim no advantage. *Status.* The trained agent models (SFT, multi-turn GRPO and DPO) and the comparisons T1–T8 have not been run; they are described as planned work and no claim about their performance is made.

9 Release and reproducibility

The product, the research lane (`ml/`) and this paper live in one repository [21]. Every run writes its configuration, data hashes, git revision and final metrics to a JSON file under `ml/results/`; the quantum study writes `ml/quantum/results.json` with instance hashes, backend, transpiled depth, shots and package versions. The paper's tables are generated from these files and from the product's evaluation results by `paper/scripts/tables.py`, which prints *not run* for any table without a result file; no number is typed into the paper by hand. The result files used for this build are: `ml/results/personas-49c52cd4.json` (ebf59fa23839); `ml/results/corpus-1e10f80a.json` (6a92bddeeabf); `ml/results/publish-corpus.json` (ee7218567658); `ml/results/publish-personas.json` (ae2e9de28762); `ml/results/gym-selftest-30060f22.json` (91602360b4f3); `ml/quantum/results.json` (14e54a79323f). Datasets, the gym and trained

Table 15: T9 QAOA on the Aer simulator: transpiled depth, two-qubit gates, feasibility rate, best approximation ratio and probability of the optimum. Source: `ml/quantum/results.json` (sha256 14e54a79323f).

Instance	Qubits	Variant	p	Depth	2q gates	Shots	Feasible	Ratio (best)	P(opt)
qa-care-noe-care-s2-d1-v11	11	qaoa	1	22	39	4,096	0.001709	1	0.0004883
qa-care-noe-care-s2-d1-v11	11	qaoa	2	34	78	4,096	0.06982	1	0.001221
qa-care-noe-care-s2-d1-v11	11	qaoa	3	46	117	4,096	0.1348	1	0.009033
qa-care-noe-care-s2-d1-v11	11	warm-start	1	22	39	4,096	0.3821	1	0.1909
qa-care-noe-care-s2-d1-v11	11	warm-start	2	34	78	4,096	0.5352	1	0.3162
qa-care-noe-care-s2-d1-v11	11	warm-start	3	46	117	4,096	0.6155	1	0.3918
qa-care-wien-care-s2-d1-v12	12	qaoa	1	24	46	4,096	0.00415	0.6667	0
qa-care-wien-care-s2-d1-v12	12	qaoa	2	37	92	4,096	0.1191	1	0.008789
qa-care-wien-care-s2-d1-v12	12	qaoa	3	50	138	4,096	0.03979	1	0.001709
qa-care-wien-care-s2-d1-v12	12	warm-start	1	24	46	4,096	0.3652	1	0.1841
qa-care-wien-care-s2-d1-v12	12	warm-start	2	37	92	4,096	0.5056	1	0.2996
qa-care-wien-care-s2-d1-v12	12	warm-start	3	50	138	4,096	0.5713	1	0.3662
qa-care-noe-care-s3-d1-v17	17	qaoa	1	34	75	4,096	0	–	0
qa-care-noe-care-s3-d1-v17	17	qaoa	2	49	150	4,096	0.01782	1	0.0007324
qa-care-noe-care-s3-d1-v17	17	qaoa	3	65	225	4,096	0.03711	1	0.0004883
qa-care-noe-care-s3-d1-v17	17	warm-start	1	34	75	4,096	0.2065	1	0.09351
qa-care-noe-care-s3-d1-v17	17	warm-start	2	49	150	4,096	0.3301	1	0.1709
qa-care-noe-care-s3-d1-v17	17	warm-start	3	65	225	4,096	0.4146	1	0.2305
qa-care-wien-care-s3-d1-v18	18	qaoa	1	31	83	4,096	0.0002441	0.3077	0
qa-care-wien-care-s3-d1-v18	18	qaoa	2	47	166	4,096	0.02563	1	0.0007324
qa-care-wien-care-s3-d1-v18	18	qaoa	3	64	249	4,096	0.03369	1	0.001221
qa-care-wien-care-s3-d1-v18	18	warm-start	1	31	83	4,096	0.2021	1	0.08691
qa-care-wien-care-s3-d1-v18	18	warm-start	2	47	166	4,096	0.3215	1	0.1655
qa-care-wien-care-s3-d1-v18	18	warm-start	3	64	249	4,096	0.3816	1	0.2175
qa-care-wien-care-s4-d1-v24	24	qaoa	1	43	136	4,096	0	–	0
qa-care-wien-care-s4-d1-v24	24	qaoa	2	60	272	4,096	0.00293	0.8	0
qa-care-wien-care-s4-d1-v24	24	qaoa	3	83	408	4,096	0.009521	0.8889	0
qa-care-wien-care-s4-d1-v24	24	warm-start	1	43	136	4,096	0.1157	1	0.04932
qa-care-wien-care-s4-d1-v24	24	warm-start	2	60	272	4,096	0.1772	1	0.09766
qa-care-wien-care-s4-d1-v24	24	warm-start	3	83	408	4,096	0.2124	1	0.1228
qa-care-wien-care-s5-d1-v30	30	qaoa	1	50	195	4,096	0	–	0
qa-care-wien-care-s5-d1-v30	30	qaoa	2	70	390	4,096	0.0002441	0.5217	0
qa-care-wien-care-s5-d1-v30	30	qaoa	3	97	585	4,096	0	–	0
qa-care-wien-care-s5-d1-v30	30	warm-start	1	50	195	4,096	0.04492	1	0.02417
qa-care-wien-care-s5-d1-v30	30	warm-start	2	70	390	4,096	0.07959	1	0.04956
qa-care-wien-care-s5-d1-v30	30	warm-start	3	97	585	4,096	0.1028	1	0.0708
qb-care-req-e3-s4	9	qaoa	1	9	13	4,096	0.3618	1	0.1621
qb-care-req-e3-s4	9	qaoa	2	15	26	4,096	0.3914	1	0.1499
qb-care-req-e3-s4	9	qaoa	3	21	39	4,096	0.3999	1	0.1348
qb-care-req-e3-s4	9	warm-start	1	9	13	4,096	0.6711	1	0.4319

32 further rows are in the result file.

Table 16: T9 QAOA on IBM hardware (backend `ibm_marrakesh`, 4,096 shots, job `dbml92d8ft8s738v2ukg`). Source: `ml/quantum/results.json` (sha256 14e54a79323f).

Instance	Qubits	Variant	p	Depth	2q gates	Phys. qubits	Feasible	Ratio (best)	P(opt)
qa-care-noe-care-s2-d1-v11	11	qaoa	1	282	143	156	0.004395	0.6667	0
qa-care-noe-care-s2-d1-v11	11	warm-start	1	287	143	156	0.2139	1	0.07031
qa-care-wien-care-s2-d1-v12	12	qaoa	1	302	174	156	0.003662	0.5	0
qa-care-wien-care-s2-d1-v12	12	warm-start	1	309	174	156	0.1729	1	0.06274
qa-care-noe-care-s3-d1-v17	17	qaoa	1	596	356	156	0.0002441	0.01194	0
qa-care-noe-care-s3-d1-v17	17	warm-start	1	592	356	156	0.03784	1	0.0127
qb-care-req-e6-s6	16	qaoa	1	103	87	156	0.1438	1	0.02075
qb-care-req-e6-s6	16	warm-start	1	103	87	156	0.4221	1	0.2332
qb-geringes-einkommen-req-e7-s6	16	qaoa	1	148	86	156	0.2075	1	0.02319
qb-geringes-einkommen-req-e7-s6	16	warm-start	1	149	86	156	0.3972	1	0.1189

models are released on the author's Hugging Face profile with dataset and model cards as they become available; Table 17 lists the artefacts already published, as recorded in the release result files. Trained models are future work.

Table 17: Published artefacts on Hugging Face, from `ml/results/publish-*.json`.

Artefact	URL	Records	Published
JulianAT/at-sv-law	`https://huggingface.co/datasets/JulianAT/at-sv-law`	2,863	2026-11-06T04:34:58+00:00
JulianAT/at-benefits-personas	`https://huggingface.co/datasets/JulianAT/at-benefits-personas`	–	2026-11-06T04:35:27+00:00

10 Conclusion

Executable eligibility rules turn benefits navigation into a task with verifiable outcomes: the correct benefits, office, documents and minimal questions can be computed for every simulated person. We described a product that encodes these rules with citations and a five-layer medical boundary, a gym that turns them into rewards, and a training and evaluation programme for open agent models built on that gym. Result files from the research lane exist for the Austrian law corpus build, the persona generator, the gym self-test and the quantum study; their tables are generated from those files. The trained agent models (SFT, multi-turn GRPO and DPO) and the comparisons T1–T8 have not been run; they are described as planned work and no claim about their performance is made. The product's own evaluation suites are reported separately as product baselines. Whether rule-trained open models can match frontier models with engine tools while keeping the boundary intact is the question the remaining experiments are designed to answer.

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